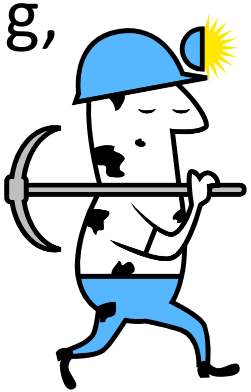
Introduction to Text Mining and Natural Language Processing

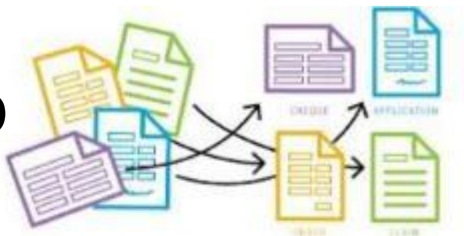
Introduction

- large volumes of data, especially in an unstructured, textual form (newspapers, blogs, company sites, discussion boards, social networks, review sites, …)
- need for automation of revealing hidden knowledge in the data => discipline known as **text mining**
- applications in marketing, security, information services, human resource management, automatic documents labeling, recommender systems, …



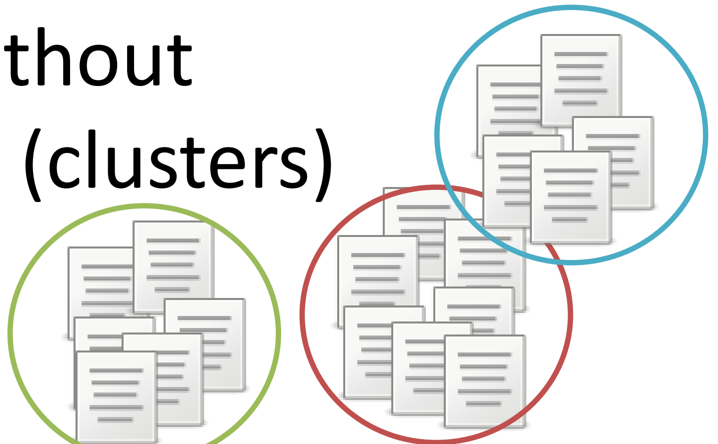
Typical tasks

- classification – assigning documents to one or more predefined classes according to training examples



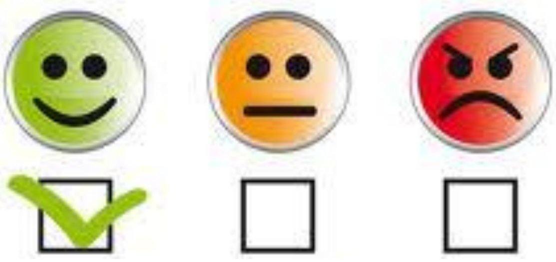
- information retrieval – retrieving documents matching a query

- clustering – grouping documents according to their similarity without prior knowledge of the groups (clusters)



Typical tasks

- summarization – selecting the most important information from one or more documents
- finding associations among document parts (words, topics, …)
- sentiment analysis and opinion mining



Typical tasks

- information extraction – finding structured information in unstructured texts (e.g., accounting data in annual reports)
- question answering – understanding questions in a natural language, extracting or formulating answers
- machine translation – computers translate texts from one language another

Typical tasks

- topic modeling – finding clusters of similar words that form the topics
- text similarity – important for retrieval, plagiarism detection, QA, etc.
- text generation – a human-like text is created based on learned patterns and language models
- dialogue systems – interactive systems keeping the context

Evolution of techniques

- rule-based (pre 1980)
- statistical methods (pre 2000)
- machine learning methods (2000-2010)
- deep learning and transformers (2010-2020)
- large language models (after 2020)

Zhang, Y. et al. A Comprehensive Survey on Automatic Text Summarization with Exploration of LLM-Based Methods. arXiv:2403.02901v2, 2025.
Chen Feng, Yifan Li, Zhaoda Chen, and Longxing Guo. 2025. The Evolution and Breakthrough of Natural Language Processing: The Revolution from Rules to Deep Learning. In Proceedings of the 2024 5th International Conference on Computer Science and Management Technology (ICCSMT '24). Association for Computing Machinery, New York, NY, USA, 307–311. https://doi.org/10.1145/3708036.3708089

Selected typical tasks

More details

Summarization

- producing a new text from *one or more* texts
- the new text conveys the most *important information* from the original text(s) and it is significantly *shorter*
- the new text can have the form of a set of key phrases, set of most important sentences, a linguistically correct abstract, …

Summarization

- basic approaches
  - extraction – identifying and reproducing most important parts
  - abstraction – producing most important parts in a new way
  - fusion – combining extracted document parts
  - compression – throwing out unimportant sections

Summarization – extraction

- early attempts focused on
  - extracting sentences containing the most frequent words (considering also a distance between the words)
  - position of sentences in paragraphs (the most informative is often at the beginning)
  - sentences containing some of the predefined words
  - parts defining the document structure (e.g., title, headings)

Summarization – extraction

- further research focused on using machine learning using features such as
  - number of words
  - first occurrence
  - frequency
  - length in characters
  - presence of a part of speech (verb, adjective, …)
- supervised approaches use existing data, such as newspaper articles, web pages, etc. with available abstracts or keywords (keyphrases)

Summarization – abstraction

- new text is generated based on a longer text
- can be characterized as ’top-down’ – they look for a set of pre-defined information types to include in the summary (e.g., an earthquake frame may contain slots for location, earthquake magnitude, number of casualties, etc.)
- the desired pieces of information are located and filled in

Multi-document summarization

- major problems
  - identifying and coping with redundancy
  - recognizing novelty, contradictions
  - ensuring logical coherence
  - ensuring completeness

Sentiment analysis

- not only objective information is important
- sentiment analysis (or opinion mining) focuses on sentiments, evaluations, attitudes, and emotions
- applications: product ranking, predicting sales performance, box-office revenues for movies, election results, stock market, and many others

Opinion definition

- an opinion is a quintuple (e_i, a_{ij}, s_{ijkl}, h_k, t_l)
  - e_i is the name of an entity
  - a_{ij} is an aspect of e_i
  - s_{ijkl} is the sentiment (positive, negative, or neutral, or expressed with different strength/intensity levels, e.g., 1 to 5) on aspect a_{ij} of entity e_i (also emotions like anger, joy, fear, etc. might be identified)
  - h_k is the opinion holder
  - t_l is the time when the opinion is expressed by h_k
- sentiment analysis objective – to identify all components
- supervised task

Liu, B. Sentiment Analysis and Opinion Mining. Morgan & Claypool Publishers, 2012.

Levels of analysis

- sentiment might be identified at
  - document/sentence level – overall sentiment for the entire piece of text
  - entity/aspect level – a sentiment is related to a target (*"Although the service is not that great, I still love this restaurant."*)
- two types of sentiment
  - regular opinions (*"Coke tastes very good"*)
  - comparative opinions (*"Coke tastes better than Pepsi"*)

Using sentiment lexicons

- the most straightforward method
  - words: positive (good, wonderful, amazing, …) and negative (bad, poor, terrible, …)
  - phrases: e.g., a cold fish (a person who does not seem very friendly and does not show their emotions)
- problems
  - *"This camera sucks"* vs. *"This vacuum cleaner really sucks"*
  - *"Can you tell me which Sony camera is good*?"
  - *"What a great car! It stopped working in two days."*
  - *"This washer uses a lot of water."*

Machine translation

- Computer Aided Translation (CAT)
  - uses software tools to facilitate the translation process, spell checkers, terminologies, concordances, translation memories tools etc.
  - help a human translator
- Machine translation
  - performs the translation itself

Machine Translation types

- rule-based
  - original form based on linguistic information and dictionaries
- example-based
  - based on analogy with previous translations (a segment is translated according to already translated sub-segment parts)
- statistical
  - the most popular MT branch some time ago
  - rules from huge bilingual content (training set) are inferred (using statistics)
  - the more training data is used, the better the results are
- hybrid
  - leverages the strengths of statistical and rule-based translation methodologies
- neural
  - are based on deep neural networks, provide SoTA results today

Information extraction

- finding structured information in unstructured data
- the desired concepts from a domain (entities, relationships) are known and IE focuses on identifying them in texts
- the output is like database records containing templates filled during an IE process
- different from complete natural language understanding => shallow linguistic analysis is usually enough

IE problems

- one fact might be expressed in many different ways
- one fact might spread across more sentences, documents, or repositories
- some information is implicit

IE tasks

- named entity recognition
  - identification and classification of predefined types of entities, such as organizations, persons, places, numerical and currency expressions, temporal expressions, etc.
- co-reference resolution
  - identification of multiple (co-referring) mentions of the same entity,
    - named: “EU” and “European Union”
    - pronominal: “John bought food. But he forgot to buy drinks.”
    - nominal: “Microsoft revealed its earnings. The company also unveiled future plans.”
    - implicit: “Petr přišel pozdě. (on, Petr) Přinesl pivo.”

IE tasks

- relation extraction
  - identification of predefined relationships
    - “Steve Jobs worked for Apple” => EmployeeOf(Steve Jobs, Apple)
    - “Mr. Smith gave a talk at the conference in New York” => LocatedIn(Smith, New York)
    - “Listed broadcaster TVN said its parent company, ITI Holdings, is considering various options for the potential sale” => SubsidiaryOf(TVN, ITI Holding)
    - etc.

IE tasks

- Event Extraction
  - identifying events and finding *who did what to whom, when, where, through what methods (instruments), and why*
  - *“Masked gunmen armed with assault rifles and grenades attacked a wedding party in mainly Kurdish southeast Turkey, killing at least 44 people.”* => perpetrators (*masked gunmen*), victims (*people*), number of killed/injured (*at least 44*), weapons and means used (*rifles and grenades*), and location (*southeast Turkey*)

Open Information Extraction

- does not require the knowledge of relations of interest (the type of extracted information is not specified in advance)
- all possible relations are found and are then available, e.g., for querying
- based on the fact that that most relations (in English) can be characterized by a set of several lexicosyntactic patterns, e.g.,
  - *Noun phrase, Verb, Noun Phrase – e.g., Graham Bell, invented, telephone*
  - *Subject, Verb, Complement* – e.g., *Albert Einstein, is, a scientist*
- uses part-of-speech tags and close word classes (e.g., pronouns, conjunctions), or clause types

Open Information Extraction

- problems
  - incoherent extractions – have no meaningful interpretation
  - uninformative extractions: “Faust made a deal with the devil” => (Faust, made, a deal) instead of (Faust, made a deal with, the devil)
  - overly-specific relation phrases: “is offering only modest greenhouse gas reduction targets at”
- solutions – better specification of linguistic constraints

Question answering

- provides information containing answers to user questions
- related to human language understanding (semantic analysis)
- in early phases limited only to some narrow domains
- can use the IE output (structured databases) for finding answers

Question answering

- NLP based
  - a query is thoroughly analysed and converted into a formal representation (logic, semantic networks, …)
  - the answer is found using a world model (e.g., an ontology)
  - ensures the most reliable answers
- IR based
  - focus on fact retrieval from a large text corpus
  - employ shallow or deep NLP techniques
  - are considered language and domain independent
- template based
  - question templates are matched against queries
  - after a match is found, an answer in a structured database is found
  - templates are typically created manually

Question answering

- closed-domain QA
  - focuses on a specific domain
  - can use knowledge for the given domain
- open-domain QA
  - any type of question about anything
  - generally needs more data

Topic modeling

- topic = probability distribution over words (e.g., topic “sport” assigns high probabilities to words “football”, “match”, “score”, etc.)
- document = mixture of topics with different probabilities
- topic modeling – unsupervised task discovering topics (themes) in document collections, i.e., latent structures that explain patterns of word co-occurrence or semantic similarity across documents

Topic modeling – methods

- probabilistic models
  - assume documents were generated with the probabilities of topics in documents and words in topics
  - maximize the likelihood = find the parameters that make the observable data most probable
  - e.g., Probabilistic Latent Semantic Analysis, Latent Dirichlet Allocation

Topic modeling – methods

- matrix factorization methods
  - a matrix representing the corpus is decomposed (approximated) into a product of two matrices (document-topic and topic-word matrices)
  - minimize the reconstruction error
  - e.g., Non-negative Matrix Factorization, Latent Semantic Analysis

Topic modeling – methods

- clustering-based
  - documents or their pieces are clustered according to their similarity
  - cluster = topic
  - usually does not allow finding multiple topics in one document (hard clustering)

Data mining vs. Text (data) mining

- some of the tasks are quite similar to the ones of data mining
- data mining
  - a well-developed branch of science
  - focuses on finding interesting patterns and hidden knowledge in large amounts of data
  - involves the following common tasks: anomaly detection, association rule learning, clustering, classification, regression, summarization
  - works with highly structured data (tabular format)

Data mining vs. Text mining

- text mining
  - focuses on tasks similar to those in DM – why not use the same methods?
  - textual data is unstructured and complicated for understanding (grammar, several meanings of a word, order of words, dictionary size, …)
  - good news – it is possible
  - but, the documents must be converted to a representation suitable for a selected DM algorithm

Data mining vs. Text mining

- text mining is related to disciplines such as statistics, artificial intelligence, and machine learning (like data mining), but also to natural language processing, linguistics, speech recognition and generation, …

Specifics of text mining

- large input space – many potential examples, huge amounts of words and their combinations
- sparse vectors representing the documents (one-hot representation) – the number of features in one document is small compared to the number of all features
- little training data – because of the demandingness of the labeling process
- noise – spelling errors, typos, wrong grammar, etc., typical for natural languages, also incorrect labeling

Specifics of text mining

- a small fraction of the content is often relevant
- the distribution of the probability with which the words appear in documents is strongly skewed

Specifics of text mining

- computational efficiency – the complexity of algorithms depends on the number of data items and attributes (often not linearly); it is necessary to develop procedures able to handle large numbers of features
- it is difficult to predict what algorithm will provide the best results, one algorithm might work for one domain and not for another

Text Mining Process

- defining the problem
- collecting the necessary data
- (defining features)
- analyzing the data (with models)
- evaluating and interpreting the results

Machine Learning for Text Mining

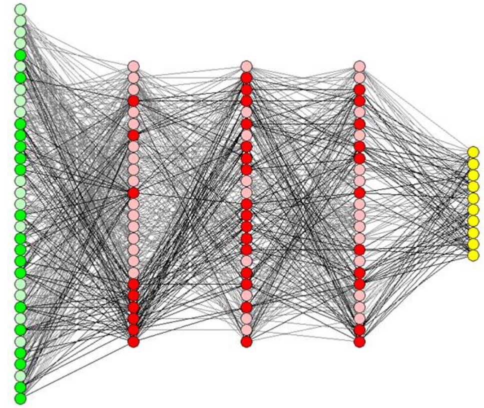
- it is not possible to cover all possible problems (inputs, situations) in advance
- past experience can be used to solve problems in the future
- machines need the ability to learn – a model to be applied in the future (in unknown situations) is trained (learned) as a generalization of past experience represented by data

Machine Learning for Text Mining

Machine Learning for Text Mining

available documents
(training data,
experience)

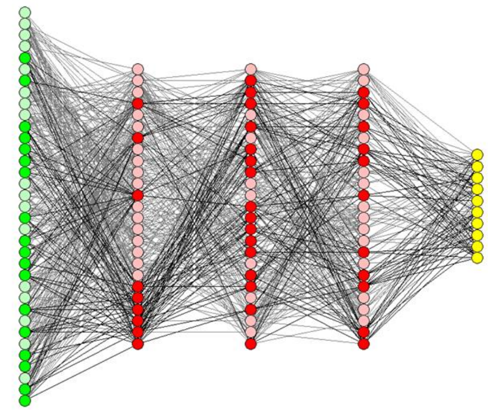
Machine Learning for Text Mining



learning (training, generalization, induction, model creation)

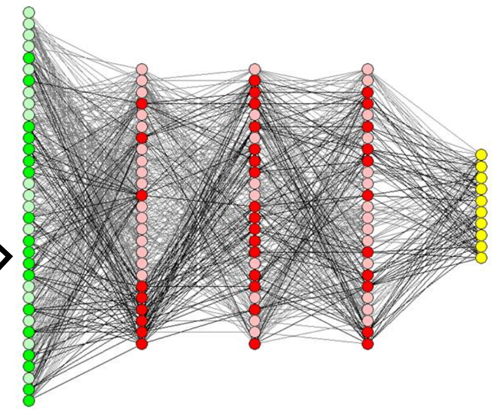
available documents (training data, experience)

Machine Learning for Text Mining



new (previously
unseen) problem

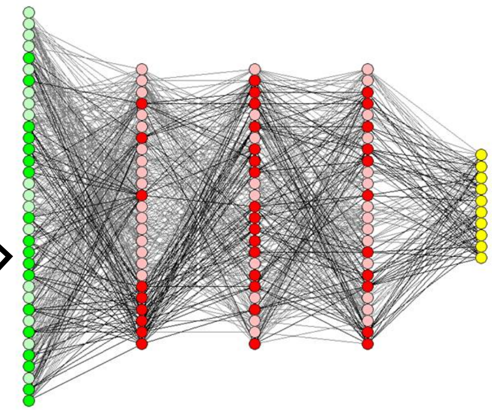
Machine Learning for Text Mining



using the
learned
model

new (previously
unseen) problem

Machine Learning for Text Mining



using the
learned
model

new (previously
unseen) problem

an answer

Machine Learning

- supervised – availability of "correct answers" (target variables, labels)
- unsupervised – no correct answers are available
- semi-supervised – a few labeled examples contain initial information for learning from many unlabeled examples